

RL-TUNED SYNTHETIC MARINE RADAR FOR SIM-TO-REAL TRANSFER

ABSTRACT

We propose a deep reinforcement learning method that adaptively tunes synthetic marine radar generation parameters to maximize downstream performance on real radar logs. The method is environment-conditioned (lake/river/coast) and targets robust sim-to-real transfer under unseen routes and clutter conditions.

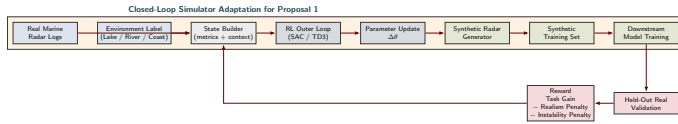


FIGURE 1: Proposal 1 closed-loop method overview.

Method Family	Real-Feedback Adaptive	Env-Type Aware	Sequential Policy Control	Physical Bounds	Task-Reward Optimized
Fixed Domain Randomization (Tobin, Peng)	✗	✗	✗	✗	✗
SimOpt / BayesSim	✓	✗	✗	✗	✗
BayRn / ADR / CDR	✓	✗	✗	✗	✗
Radar Calibration (Digital Twin style)	✓	✗	✗	✓	✗
Proposed Method (Proposal 1)	✓	✓	✓	✓	✓

FIGURE 2: Related-work capability comparison.

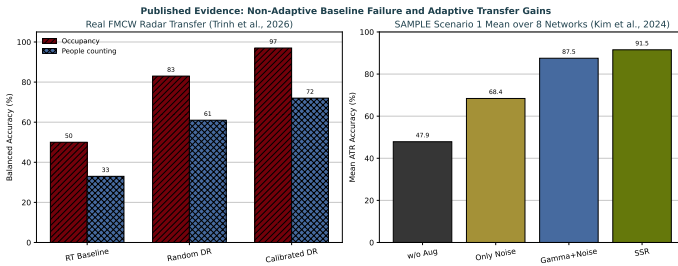


FIGURE 3: Published quantitative evidence motivating Proposal 1.

1. INTRODUCTION

Reliable marine autonomy is difficult because real radar data collection is expensive, sparse across conditions, and often misaligned with target deployment environments. Synthetic marine

radar generation is an attractive alternative, but fixed simulator settings produce a persistent sim-to-real gap that limits downstream detection and mapping performance.

This proposal studies whether a reinforcement learning policy can adaptively tune synthetic radar generator parameters to maximize real-data validation performance across environment types (lake, river, coast). The central hypothesis is that environment-conditioned simulator control can outperform fixed or random domain randomization and reduce transfer error under unseen routes and weather/clutter conditions.

The approach is grounded in prior work on domain randomization and adaptive sim-to-real transfer [1–13]. However, existing methods either rely on fixed/random randomization, non-RL adaptive tuning, or calibration-centric transfer that is not explicitly conditioned on maritime environment type. Our proposal targets this gap with an environment-aware RL outer loop that directly optimizes real-data downstream task performance, as summarized in Fig. 1 and contrasted with prior work in Fig. 2.

2. APPROACH

We formulate synthetic radar tuning as a bi-level optimization problem. In the inner loop, a downstream model is trained on synthetic radar generated with parameter vector θ . In the outer loop, an RL policy observes the current generator state, environment-type label, and recent real-validation metrics, then outputs updates to θ .

The action space controls key simulator factors including clutter intensity, noise floor, attenuation, speckle, and target reflectivity variation. The reward is defined as real-data task improvement minus realism/stability penalties to discourage unphysical parameter exploitation. We use Soft Actor-Critic (SAC) as the primary outer-loop optimizer due to sample efficiency and continuous-control support, with TD3 as a backup. A non-RL Bayesian optimization tuner is included as a strong baseline.

Relative to existing adaptive randomization methods (SimOpt, BayesSim, BayRn) [3–5], our method makes three concrete changes: (1) sequential policy-based control of simulator updates rather than one-step or purely inference-driven adaptation, (2) explicit conditioning on maritime environment type (‘lake’, ‘river’, ‘coast’), and (3) physically bounded

action constraints with realism penalties to prevent simulator exploitation. Relative to radar calibration-oriented sim-to-real pipelines [11], the optimization target is not just simulation fidelity but downstream real-task performance under domain holdout.

3. EVALUATION

Evaluation uses real marine radar logs with explicit environment labels and strict split protocols by route/location to prevent leakage. We compare: (1) fixed synthetic defaults, (2) uniform random domain randomization, (3) Bayesian optimization tuning, (4) RL tuning without environment conditioning, and (5) RL tuning with conditioning.

Primary metrics are downstream real-data task score and relative sim-to-real gap reduction. Secondary metrics include cross-domain generalization (train on two environment types, test on held-out type), unseen-location robustness, seed variance, and performance under low-SNR/high-clutter stress conditions. We will report both best-case performance and sample-efficiency measured by real-validation calls required to reach target accuracy.

To explicitly support the novelty claims, we add claim-linked evaluations: (a) sequential adaptation benefit by comparing RL vs BayRn under matched real-evaluation budgets [5]; (b) environment semantics benefit by comparing conditioned vs unconditioned RL policies; and (c) plausibility benefit by reporting parameter-bound violations and instability with/without realism regularization. These analyses directly test how our method improves over existing randomization and simulator-adaptation approaches [1–4, 6, 7].

Published results already show clear failure modes for non-adaptive baselines (Fig. 3). In real FMCW radar transfer, simulation-only baselines can collapse to near-chance balanced accuracy (about 50% for occupancy and 33% for people-counting), while calibrated adaptation substantially improves performance (about 97% and 72%, respectively) [11]. In synthetic-to-measured SAR transfer, synthetic-only and simple-noise baselines are consistently weaker than stronger adaptation strategies across multiple architectures [10]. These patterns motivate the need for a more structured, environment-aware adaptation controller rather than fixed randomization.

4. CONCLUSION

This project targets a practical and research-relevant problem: improving sim-to-real transfer for marine radar perception through adaptive synthetic data generation. The expected contribution is an RL-controlled simulator tuning framework that is environment-aware, benchmarked against strong non-RL alternatives, and evaluated under realistic transfer stress tests. If full RL tuning is unstable, a contextual-bandit or Bayesian fallback will still provide a complete, meaningful project outcome.

REFERENCES

[1] Tobin, Josh, Fong, Rachel, Ray, Alex, Schneider, Jonas, Zaremba, Wojciech and Abbeel, Pieter. “Domain Randomization for Transferring Deep Neural Networks from Simulation to the Real World.” *arXiv preprint arXiv:1703.06907* (2017).

[2] Peng, Xue Bin, Andrychowicz, Marcin, Zaremba, Wojciech and Abbeel, Pieter. “Sim-to-Real Transfer of Robotic Control with Dynamics Randomization.” *ICRA* (2018).

[3] Chebotar, Yevgen, Handa, Ankur, Makoviychuk, Viktor, Macklin, Miles, Issac, Jan, Ratliff, Nathan and Fox, Dieter. “Closing the Sim-to-Real Loop: Adapting Simulation Randomization with Real World Experience.” *arXiv preprint arXiv:1810.05687* (2018).

[4] Ramos, Fabio, Possas, Rafael and Fox, Dieter. “BayesSim: Adaptive Domain Randomization via Probabilistic Inference for Robotics Simulators.” *arXiv preprint arXiv:1906.01728* (2019).

[5] Muratore, Fabio, Gienger, Michael and Peters, Jan. “Data-Efficient Domain Randomization with Bayesian Optimization.” *IEEE Robotics and Automation Letters* (2020).

[6] OpenAI et al. “Solving Rubik’s Cube with a Robot Hand.” *arXiv preprint arXiv:1910.07113* (2019).

[7] Josifovski, Martin, Cruz, Francisco, Boedecker, Joschka and Wang, Jun. “Continual Domain Randomization.” *arXiv preprint arXiv:2403.12193* (2024).

[8] Ho, Jonathan et al. “RAP: Learning to Simulate Robust and Accurate Physics for 3D Manipulation.” *arXiv preprint arXiv:2005.07141* (2020).

[9] Zhao, Wenhao et al. “Sim-to-Real Transfer in Deep Reinforcement Learning for Robotics: A Survey.” *arXiv preprint arXiv:2009.13303* (2020).

[10] Kim, Wonseok et al. “Soft Segmented Randomization: Enhancing Domain Generalization in SAR-ATR for Synthetic-to-Measured Applications.” *arXiv preprint arXiv:2409.14060* (2024).

[11] Trinh, Dinh, Deshmukh, Harshal, Herath, Sanjana, Waslander, Steven and Trinh, Hieu. “A Physics-Informed Digital Twin Framework for Calibrated Sim-to-Real FMCW Radar Occupancy Estimation.” *arXiv preprint arXiv:2601.17871* (2026).

[12] Zust, Luka et al. “LaRS: A Diverse Panoptic Maritime Obstacle Detection Dataset and Benchmark.” *arXiv preprint arXiv:2308.09618* (2023).

[13] Choi, Jaehyun et al. “PoLaRIS Dataset: A Maritime Object Detection and Tracking Dataset in Pohang Canal.” *arXiv preprint arXiv:2412.06192* (2024).